

COMP550

TRENDING TOPICS IN DEEP LEARNING

Aykut Erdem

Department of Computer of Engineering
KUIS Artificial Intelligence Center



KOÇ
UNIVERSITY

Fall 2026

Nine questions for the seminar

- | | | |
|-----------|---|---|
| 01 | Post-training and RL | How do feedback and rewards change behavior? |
| 02 | Efficient adaptation and compression | How much can we learn or deploy with fewer resources, or specialize to other domains? |
| 03 | Reasoning and inference-time computation | When/how does spending more at inference help? |
| 04 | Unified multimodal models | Can one model both interpret and create multiple modalities? |
| 05 | World models and physical intelligence | Can a model predict and act in the physical world? |
| 06 | Retrieval and long-term memory | What should persist, and how is it retrieved? |
| 07 | Tool-using agents | How should the model act outside its parameters? |
| 08 | Interpretability and model steering | Can we identify and steer internal computations? |
| 09 | Evaluation and benchmarking | What evidence would justify a capability claim? |

Post-training and RL

01

How does a next-token predictor become a helpful assistant —
and then a reasoner?

Instruction tuning

RLHF

DPO

RL with verifiable rewards

GRPO

Reward hacking

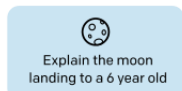
Post-training turns a predictor into a policy

THE OBJECTIVE CHANGES FROM PREDICTING OBSERVED DATA TO PRODUCING USEFUL BEHAVIOR.

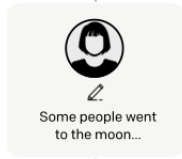
Step 1

Collect demonstration data, and train a supervised policy.

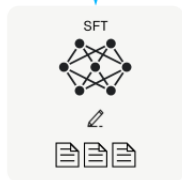
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



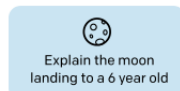
This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

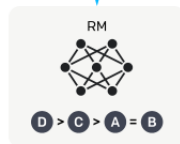
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.



The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



Pretraining

Learn broad representations and next-token prediction.

Supervised fine-tuning (SFT)

Imitate demonstrations of desired responses.

Reward optimization

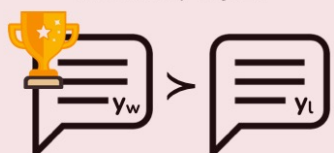
Increase the probability of responses that score well.

Feedback can drive different learning procedures

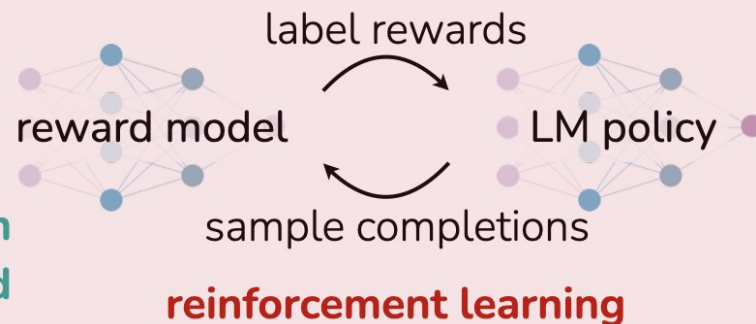
PREFERENCE OPTIMIZATION AND ONLINE RL USE FEEDBACK IN DIFFERENT WAYS.

Reinforcement Learning from Human Feedback (RLHF)

x: "write me a poem about
the history of jazz"



maximum
likelihood



Direct Preference Optimization (DPO)

x: "write me a poem about
the history of jazz"



maximum
likelihood



SFT: demonstrations

Train on a target response.
The signal says: imitate this.

DPO: preference pairs

Increase preferred-vs-rejected
odds relative to a reference
policy.

Online RL: new samples

Generate, score, update.
PPO or GRPO are examples.

What should a visual reasoning reward measure?

**A model gives the correct answer
but cites a clue that does not support it.**

An incomplete reward

Answer accuracy alone rewards the final choice—even if the evidence is wrong.

A better measurement target

Check answer correctness, visual grounding, and appropriate uncertainty separately.

Would you reward this response? What would your checker need to observe?

Optimizing a proxy can improve the score while preserving the wrong behavior.

Efficient adaptation and compression

02

How little can we train, store and compute — and still keep what the model knows?

LoRA

QLoRA

Quantization

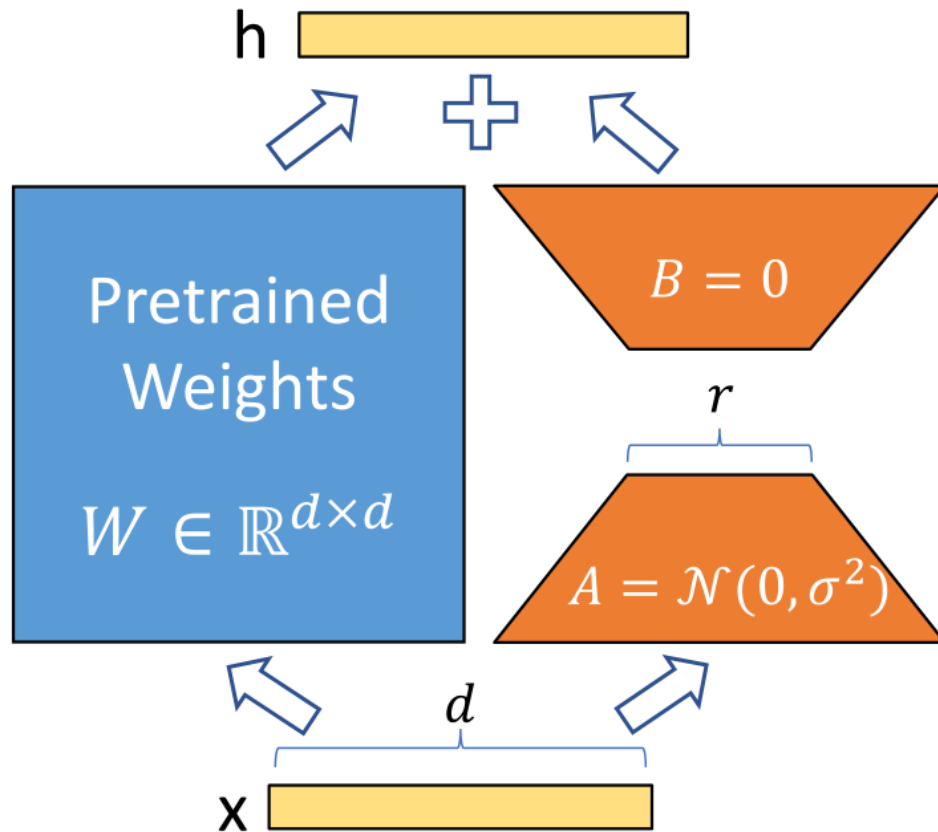
Pruning

Distillation

Mixture-of-experts

Adaptation can train a small change to the weights

LORA KEEPS THE BASE MATRIX FIXED AND LEARNS A LOW-RANK UPDATE.



$$W' = W + BA$$

W is frozen; the smaller matrices A and B are trainable.

A concrete scale comparison

4096×4096 : 16.8M parameters.

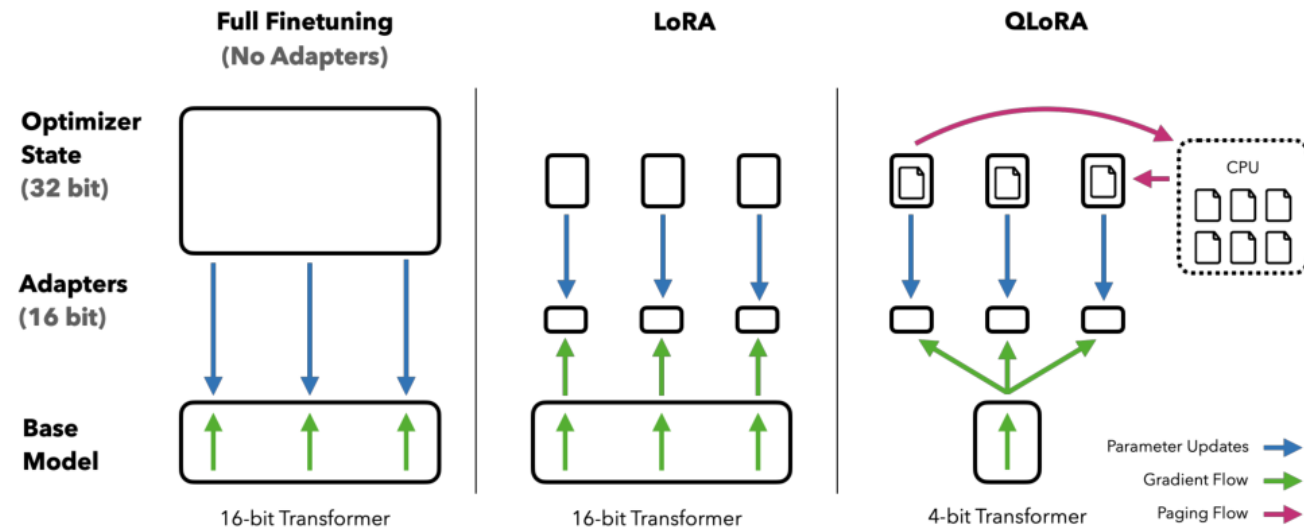
Rank 8 update: 65,536 parameters.

What this saves

Trainable parameters and optimizer state; the base weights still exist.

Adaptation and compression solve different problems

QLORA COMBINES AN ADAPTATION METHOD WITH A COMPRESSED REPRESENTATION OF THE BASE WEIGHTS.



Adaptation

Fit a new task or domain.
Examples: full tuning, LoRA.

Compression

Reduce storage or computation.
Examples: quantization, pruning, distillation.

QLoRA combines them

A frozen 4-bit base plus trainable LoRA adapters.

Visual tokens are also a resource budget

ILLUSTRATIVE VIDEO EXAMPLE: 16 FRAMES, WITH A DIFFERENT TOKEN BUDGET PER FRAME.

Representation	Visual tokens	Dense attention score entries
256 tokens per frame	4,096	16,777,216
64 tokens per frame	1,024	1,048,576

4× fewer visual tokens; 16× fewer entries in the visual-token attention matrix.

Which evidence disappears first: small text, tiny objects, or brief actions?

Unified multimodal understanding and generation

03

Can one model both understand and generate images — and does each skill help the other?

Vision encoders

Visual tokenizers

Early fusion

Autoregressive

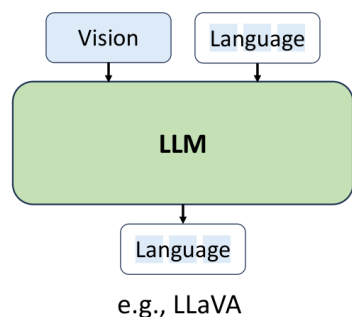
Diffusion & flow

Interleaved generation

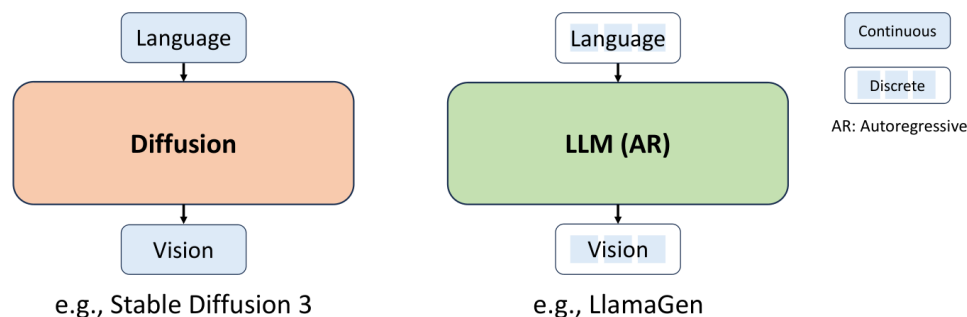
Unified models learn to understand and generate

MULTIMODALITY CONCERNS THE INPUTS AND OUTPUTS; UNIFICATION CONCERNS HOW CAPABILITIES SHARE A MODEL.

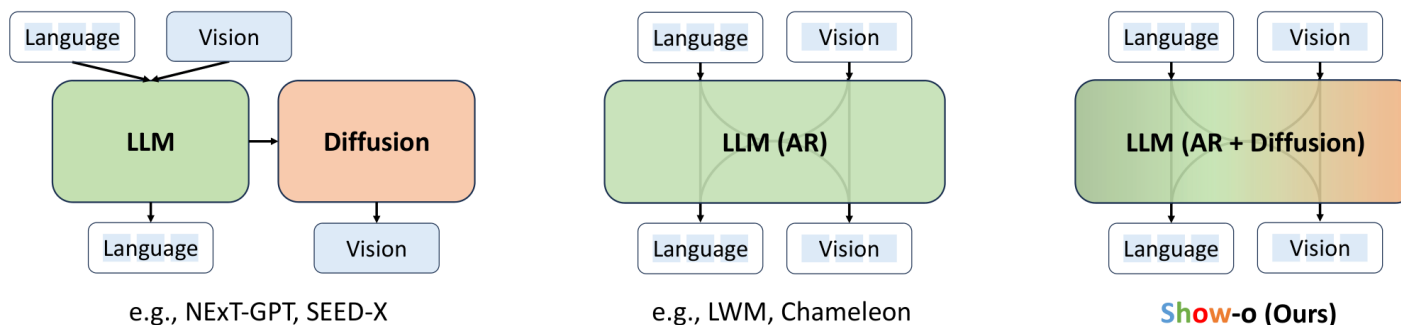
(a) Understanding Only



(b) Generation Only



(c) Unified Model (Understanding & Generation)



Understanding

Condition on an image and question; produce an answer.

Generation

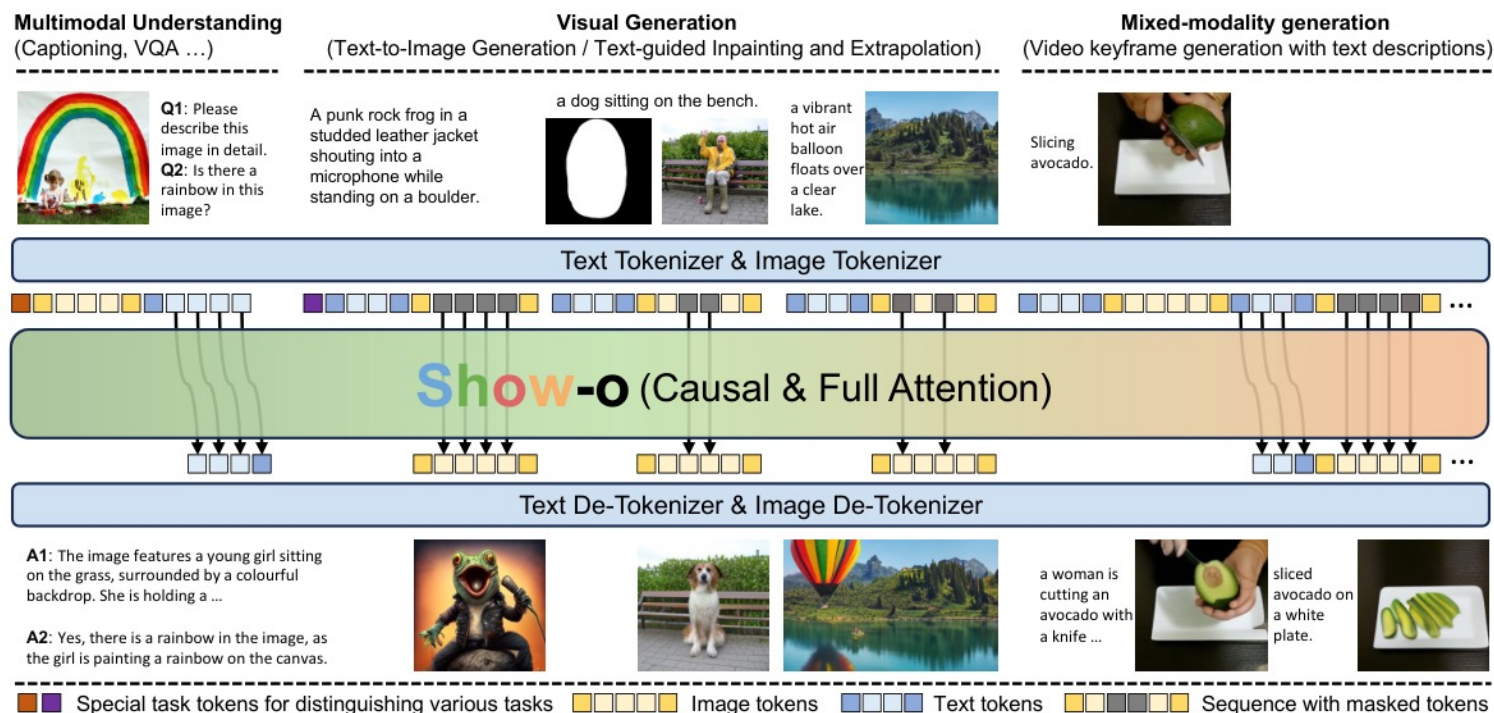
Condition on text or images; produce new visual content.

Joint capability

Share computation across directions, including interleaved tasks.

A shared backbone can use different objectives

Show-o illustrates autoregressive language modeling together with visual denoising.



Represent the modalities

Encode text and visual content into model-compatible tokens.

Choose how to predict

Text: predict the next token.
Image: iteratively recover masked tokens.

Share selectively

Balance shared features against conflicting training objectives.

Test whether the two capabilities help each other

**“Describe the scene. Make the cup blue.
Now identify what changed.”**

What should stay consistent?

Object identity, geometry, relations, and everything the edit did not target.

What would count as evidence?

Evaluate answering, editing, and cross-task consistency against matched baselines.

Does a convincing generated image show that the model understood the original?

Joint capability is a starting point; beneficial transfer requires an experiment.

Background: Xie et al. · Show-o · ICLR 2025

Reasoning and inference-time computation

04

If thinking longer helps, how should a model spend its compute at test time?

Chain of thought

Self-consistency

Best-of-N

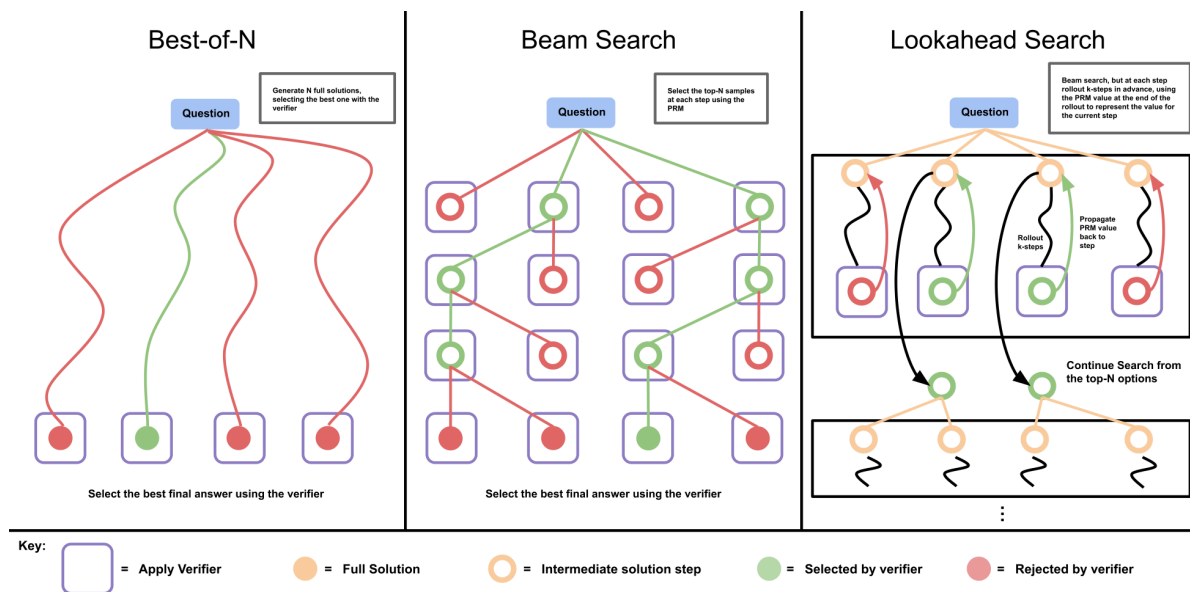
Reward models

Tree search

Thinking budgets

Inference-time compute buys additional attempts

At fixed weights, spend more computation on one problem before committing to an answer.



Generate alternatives

Sample several complete solutions or expand partial solutions.

Evaluate candidates

Use a verifier, a reward model, or agreement among answers.

Allocate the budget

Spend more on difficult or uncertain cases when that helps.

Search is only as useful as its selection rule

A MODEL MAY GENERATE A CORRECT CANDIDATE AND STILL CHOOSE THE WRONG ANSWER.

Strategy	Selection signal	Typical failure
Self-consistency	Most frequent final answer	Correlated errors win the vote
Best-of-N	Highest verifier score	The verifier prefers a wrong answer
Sequential revision	Critique and improve a candidate	Revision reinforces the initial mistake

pass@k asks whether any candidate is correct. Deployment must choose one.

Sometimes the next useful step is to look again

Did the cup fall before or after the gripper opened?

Spend on more language

Generate longer explanations of the same sampled frames.

Spend on better evidence

Retrieve the transition, inspect more frames, and compare timestamps.

If the decisive frame was never observed, what can another reasoning chain fix?

Allocate computation to the bottleneck: perception, search, or verification.

Background: Snell et al. · Scaling LLM Test-Time Compute Optimally · ICLR 2025

World models and physical intelligence

05

Can models learn how the world evolves — and use that knowledge to act in it?

Latent dynamics

JEPA

Video generation

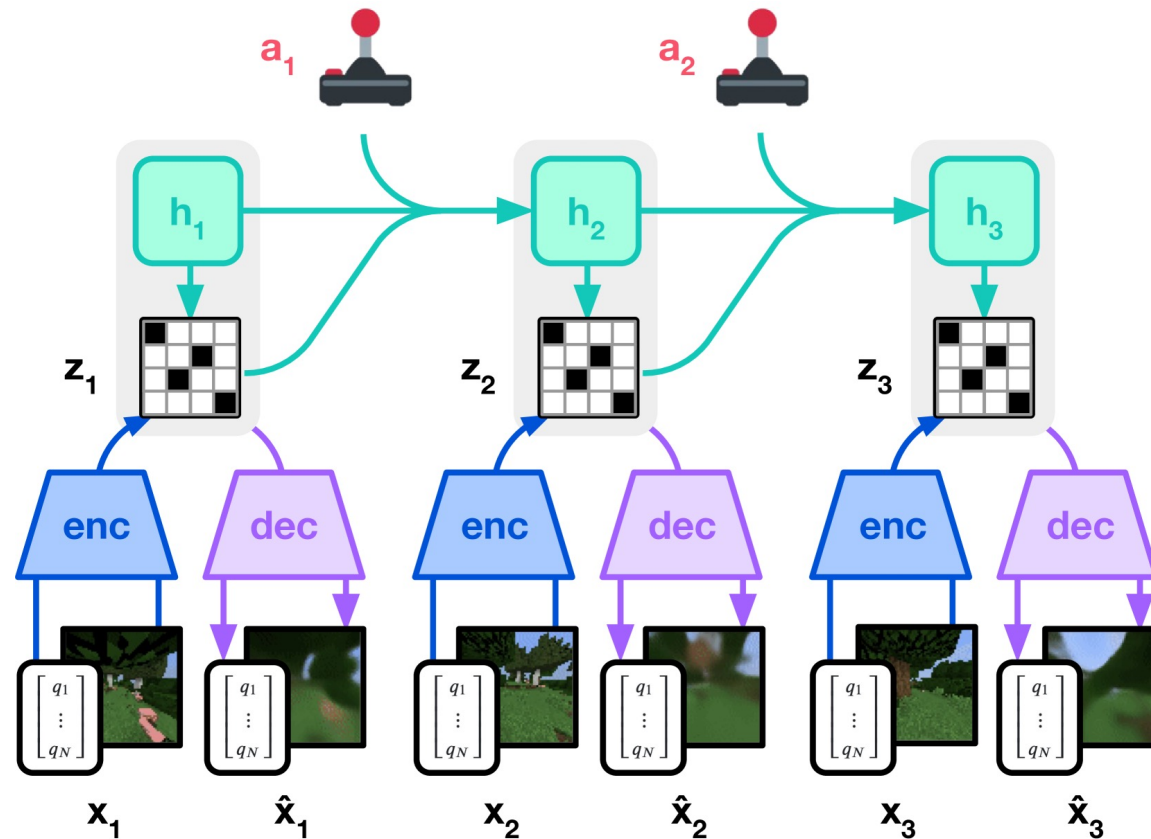
Interactive environments

Vision-language-action

Embodiment

World models predict consequences of actions

LEARN A COMPACT STATE AND HOW IT EVOLVES; USE IMAGINED FUTURES FOR PLANNING OR POLICY LEARNING.



Infer a state

Compress observations and history into a useful latent representation.

Predict a transition

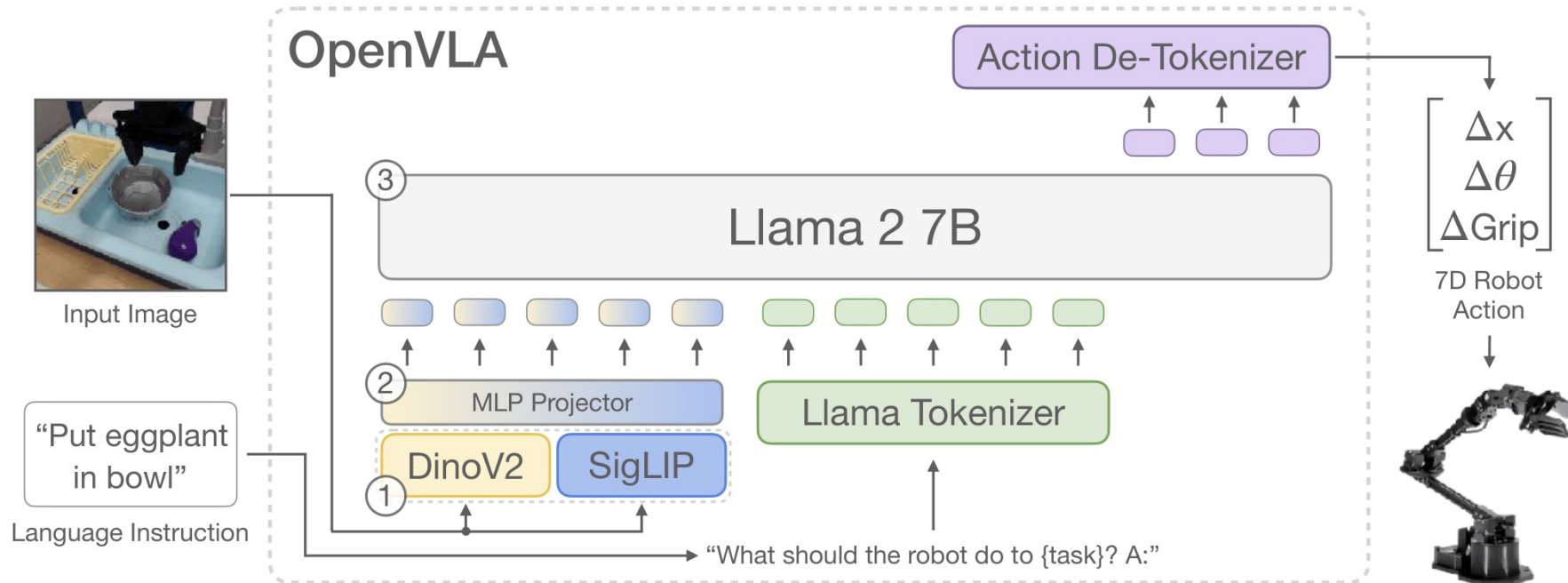
Condition future state on current state and action.

Use imagined experience

Predict outcomes or rewards to compare actions or train a policy.

Physical intelligence connects perception to action

A VISION-LANGUAGE-ACTION (VLA) POLICY MAPS OBSERVATIONS AND INSTRUCTIONS TO ROBOT ACTIONS.



VLA policy

What action should I take?
Predict action tokens or controls.

World model

What may happen if I act?
Predict future states.

A closed-loop system

Execute, observe the result, and
update the next decision.

A plausible future is not yet a physical model

**A generated clip looks realistic,
but the released cup remains suspended.**

Visual realism

Texture, lighting, and smooth motion can look convincing.

Physical and causal consistency

Contact, support, gravity, object persistence, and action consequences must agree.

How would you distinguish memorized motion from a model that supports control?

Test counterfactual actions, longer horizons, and closed-loop task success.

Long-term memory and long context 06

Where should knowledge live — in the weights, in the context, or in an external store?

Retrieval-augmented generation

KV cache

Long context

State-space models

Agent memory

Knowledge editing

"Memory" can live in several different places

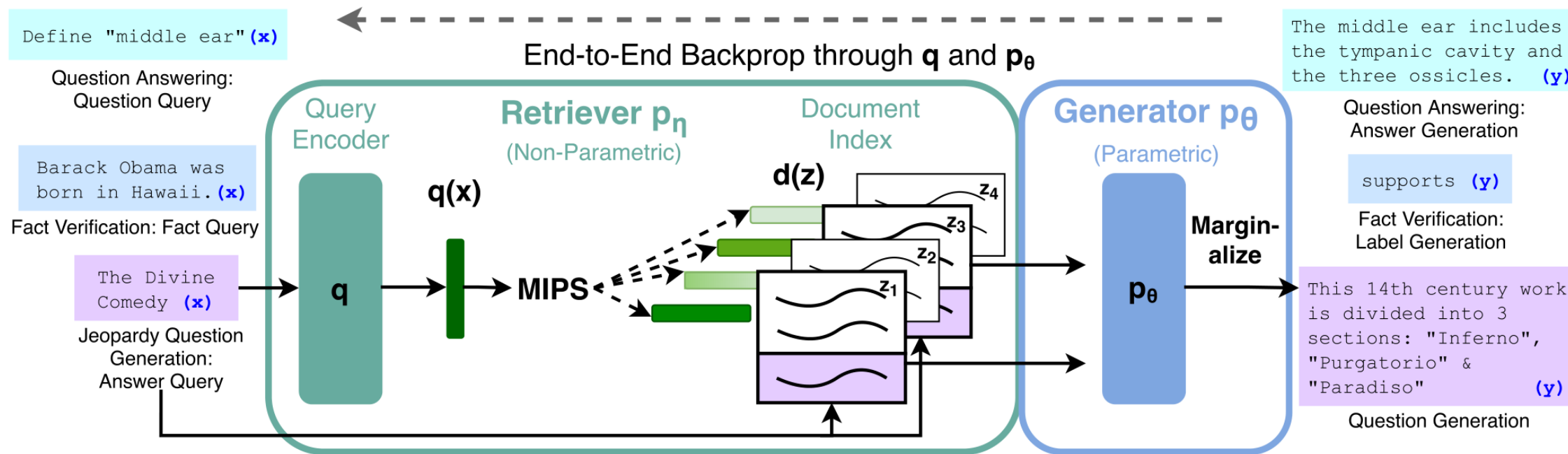
SEPARATE LEARNED KNOWLEDGE, TEMPORARY CONTEXT, AND AN EXTERNAL RECORD THAT CAN BE UPDATED.

Where it lives	How it is written	Main limitation
Model parameters	Training or weight editing	Updates are costly and hard to trace
Context + attention cache	Supply tokens during inference	Bounded context; cache is not durable memory
External memory store	Save events, documents, or summaries	Retrieval and forgetting must be designed

Long-term memory needs a write policy, a read policy, and a way to revise or forget.

Retrieval adds evidence to the current context

RAG SEPARATES FINDING POTENTIALLY RELEVANT INFORMATION FROM USING IT TO PRODUCE AN ANSWER.



Write / index

Segment content; retain source identifiers and timestamps.

Read / retrieve

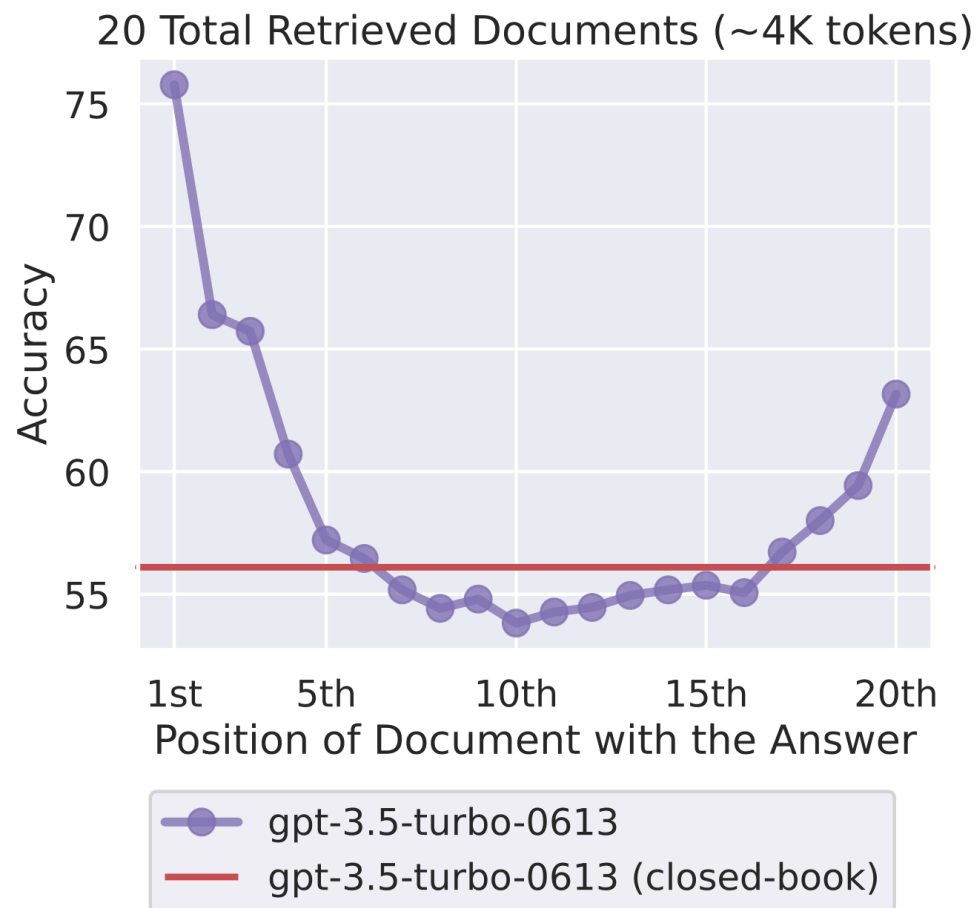
Match the query to useful records, then rerank or filter.

Use / verify

Answer from retrieved evidence; check support and recency.

A large context window is not a retrieval guarantee

THE POSITION AND FORM OF EVIDENCE CAN CHANGE WHETHER A MODEL USES IT.



Read the historical result

In this study, relevant evidence was often used less reliably in the middle.

Transfer the test, not the score

Vary position, distractors, length, and temporal order on your target model.

For long video

“Where was the cup last seen?” requires identity, recency, and provenance.

Tool-using agents

07

What changes when a model's outputs become actions with real consequences?

ReAct

Function calling

Model Context Protocol

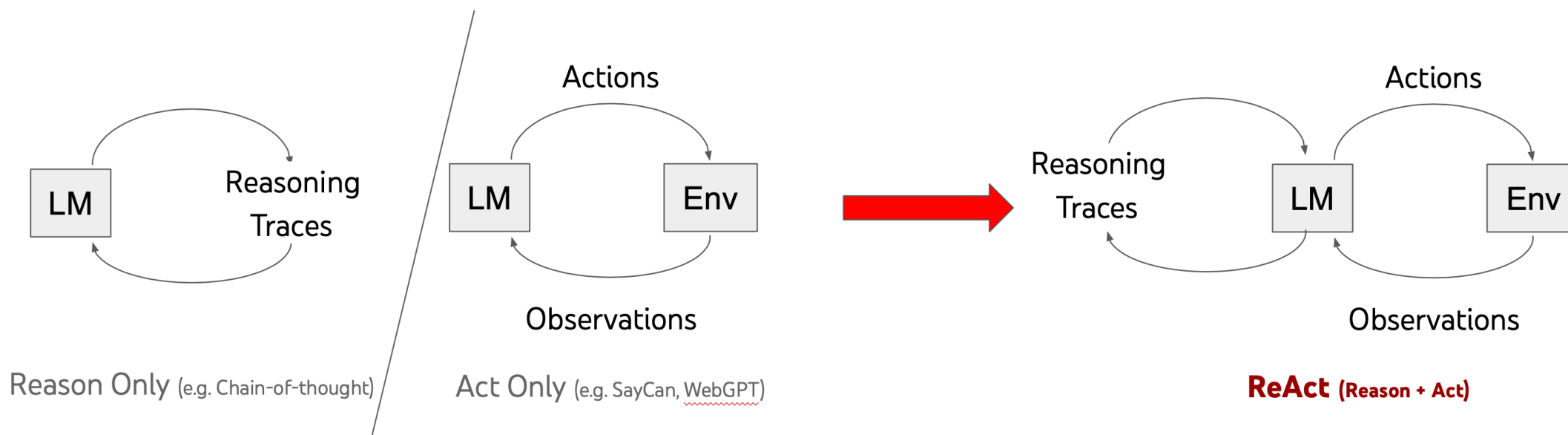
Computer use

Coding agents

Prompt injection

An agent acts, observes, and updates its plan

TOOL USE LETS A MODEL OBTAIN NEW EVIDENCE OR CHANGE AN EXTERNAL ENVIRONMENT.



Choose an action

Select a tool and provide arguments under a defined interface.

Read the observation

Use the actual tool result, including errors and uncertainty.

Continue or stop

Revise the plan, call another tool, or produce a grounded answer.

Visual tool use can make evidence inspectable

ILLUSTRATIVE TASK: READ THE SERIAL NUMBER ON A SMALL DEVICE IN A VIDEO.

Step	Tool or operation	Observation to preserve
Locate	Retrieve frames with the device	Frame IDs and timestamps
Inspect	Crop the device; run OCR	Crop coordinates and candidate text
Verify	Compare a nearby frame or another crop	Agreement or a visible ambiguity
Answer	Report the supported reading	Serial number + source; abstain if unclear

The tool trace should distinguish observed evidence from model-generated guesses.

More tool calls can amplify an early mistake

**A detector selects the wrong device.
Every later crop and OCR call follows it.**

Local success can mislead

Each call returns a valid result, but the overall task is still wrong.

Evaluate the complete loop

Track task success, evidence quality, calls, latency, and recovery from errors.

What observation should trigger replanning instead of another identical call?

Useful agents need feedback, stopping rules, and ways to recover.

Background: Yao et al. · ReAct · ICLR 2023

Mechanistic interpretability and model steering

08

Can we read — and then write — the computations inside a network?

Superposition

Sparse autoencoders

Circuits

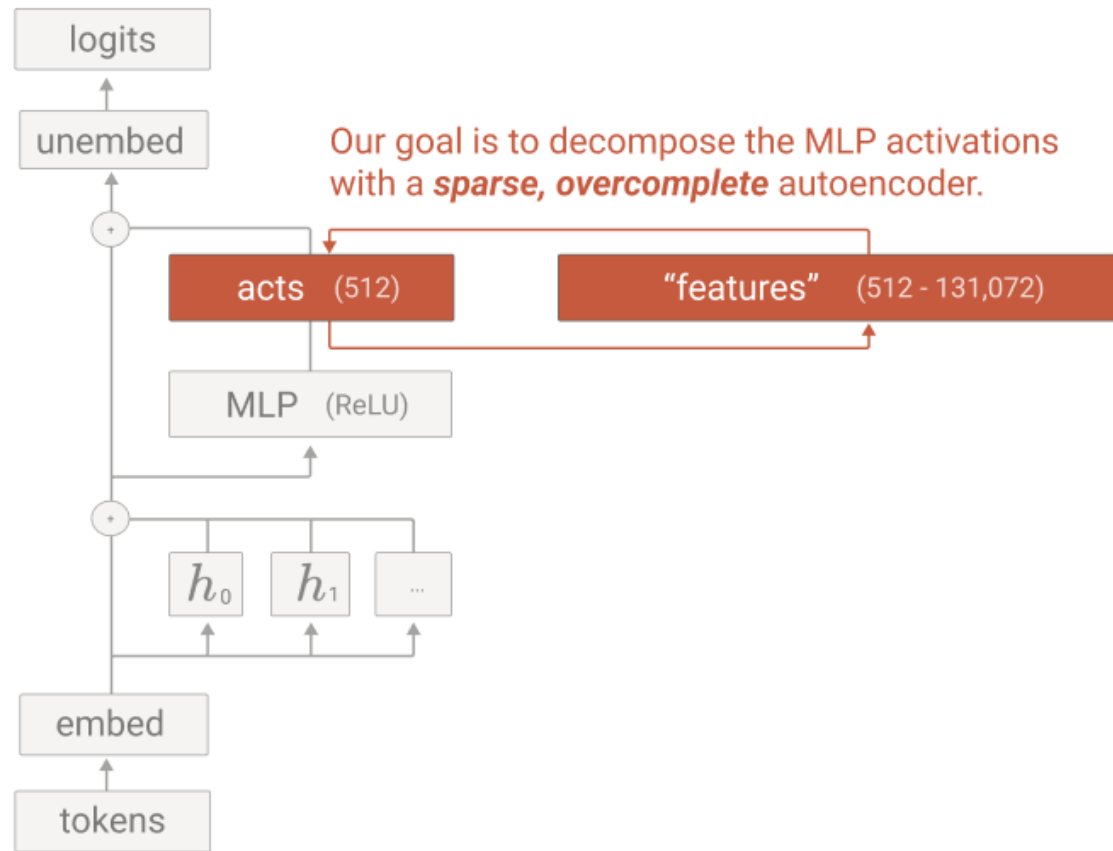
Activation patching

Steering vectors

Model editing

Mechanistic interpretability asks how a model computes

FIND INTERNAL FEATURES AND CIRCUITS, THEN TEST THEIR ROLE IN A SPECIFIC BEHAVIOR.



Distributed representations

A neuron may participate in several features; a feature may span neurons.

Sparse autoencoders (SAEs)

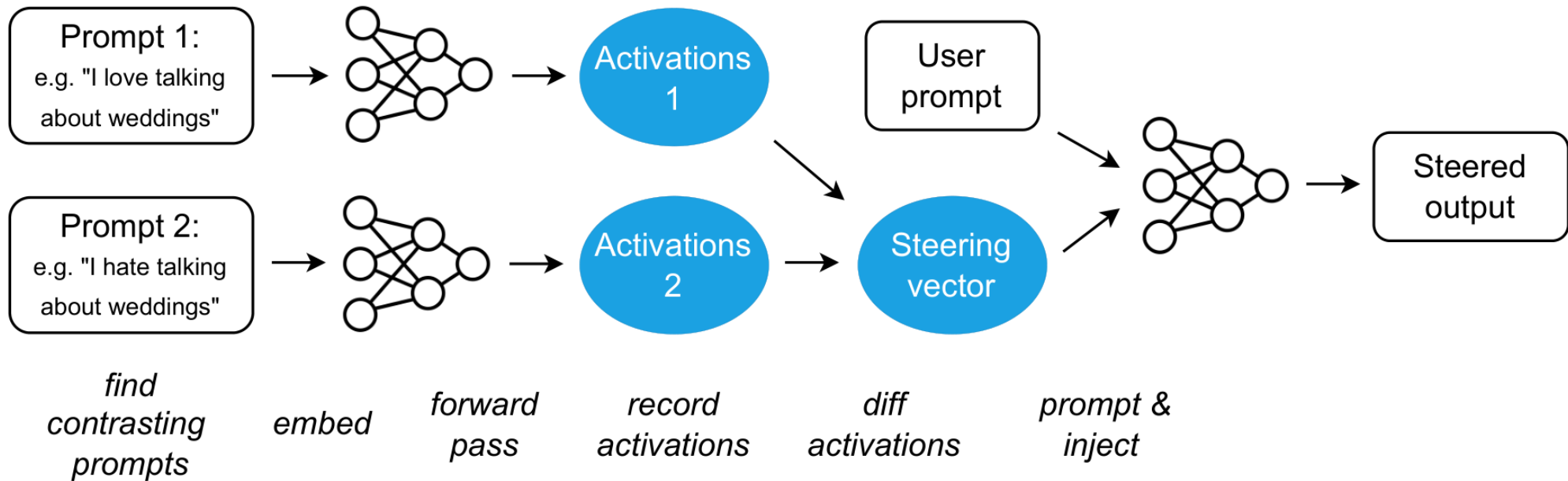
Reconstruct activations using a larger set of sparsely active features.

A description is a hypothesis

A labeled feature needs behavioral and causal validation.

Steering intervenes during the forward pass

ACTIVATION ADDITION CHANGES A HIDDEN REPRESENTATION ALONG A CHOSEN DIRECTION.



Construct a direction

Contrast activations from paired prompts or labeled examples.

Apply an intervention

$h' = h + \alpha v$
Choose a layer, position, and strength α .

Measure the trade-off

Desired behavior may improve while other capabilities degrade.

A causal claim needs a controlled intervention

ILLUSTRATIVE VLM HYPOTHESIS: AN INTERNAL COMPONENT CARRIES THE CUP'S COLOR.

Experiment	Observation	What it supports
Probe red vs. blue examples	Color is decodable from activations	Information is present
Patch activations between matched images	The answer changes toward the donor color	A causal contribution in this setting
Use control patches; test new scenes	Effect is selective and reproducible	A stronger, still bounded mechanism claim

A generated explanation, a probe, and a causal intervention are different kinds of evidence.

Evaluation and benchmarking

09

How do we know a model is better — and better at what?

Saturation

Contamination

Construct validity

LLM-as-a-judge

Arenas

Living benchmarks

A benchmark operationalizes a capability claim

A SCORE MEASURES PERFORMANCE ON SPECIFIED TASKS, DATA, AND A SCORING PROCEDURE.

Art & Design	Business	Science
Question: Among the following harmonic intervals, which one is constructed incorrectly? Options: (A) Major third <i><image 1></i> (B) Diminished fifth <i><image 2></i> <u>(C) Minor seventh <i><image 3></i></u> (D) Diminished sixth <i><image 4></i>	Question: ...The graph shown is compiled from data collected by Gallup <i><image 1></i>. Find the probability that the selected Emotional Health Index Score is between 80.5 and 82? Options: (A) 0 (B) 0.2142 <u>(C) 0.3571</u> (D) 0.5	Question: <i><image 1></i> The region bounded by the graph as shown above. Choose an integral expression that can be used to find the area of R. Options: <u>(A) $\int_0^{1.5} [f(x) - g(x)] dx$</u> (B) $\int_0^{1.5} [g(x) - f(x)] dx$ (C) $\int_0^2 [f(x) - g(x)] dx$ (D) $\int_0^2 [g(x) - x(x)] dx$
Subject: Music; Subfield: Music; Image Type: Sheet Music; Difficulty: Medium	Subject: Marketing; Subfield: Market Research; Image Type: Plots and Charts; Difficulty: Medium	Subject: Math; Subfield: Calculus; Image Type: Mathematical Notations; Difficulty: Easy
Health & Medicine	Humanities & Social Science	Tech & Engineering
Question: You are shown subtraction <i><image 1></i>, T2 weighted <i><image 2></i> and T1 weighted axial <i><image 3></i> from a screening breast MRI. What is the etiology of the finding in the left breast? Options: (A) Susceptibility artifact (B) Hematoma <u>(C) Fat necrosis</u> (D) Silicone granuloma	Question: In the political cartoon, the United States is seen as fulfilling which of the following roles? <i><image 1></i> Option: (A) Oppressor (B) Imperialist <u>(C) Savior</u> (D) Isolationist	Question: Find the VCE for the circuit shown in <i><image 1></i>. Neglect VBE Answer: 3.75 Explanation: ...$I_E = [(V_{EE}) / (R_E)] = [(5 \text{ V}) / (4 \text{ k-ohm})] = 1.25 \text{ mA}$; $V_{CE} = V_{CC} - I_{ERL} = 10 \text{ V} - (1.25 \text{ mA}) 5 \text{ k-ohm}$; $V_{CE} = 10 \text{ V} - 6.25 \text{ V} = 3.75 \text{ V}$
Subject: Clinical Medicine; Subfield: Clinical Radiology; Image Type: Body Scans: MRI, CT.; Difficulty: Hard	Subject: History; Subfield: Modern History; Image Type: Comics and Cartoons; Difficulty: Easy	Subject: Electronics; Subfield: Analog electronics; Image Type: Diagrams; Difficulty: Hard

Define the construct

Perception? Domain knowledge? Spatial or temporal reasoning?

Design the tasks

Choose inputs and questions that require the intended capability.

Test alternative explanations

Could text-only clues, memorization, or shortcuts solve it?

The same average can hide very different failures

ILLUSTRATIVE BENCHMARK WITH 1,000 EXAMPLES: 900 COMMON CASES AND 100 RARE CASES.

Model	Common cases (900)	Rare cases (100)	Overall accuracy
A	95%	45%	90%
B	90%	90%	90%

Model A: 855 + 45 correct. Model B: 810 + 90 correct.

Which model would you deploy where the rare cases matter most?

Benchmark the claim, the protocol, and the cost

A STRONG EVALUATION MAKES COMPETING EXPLANATIONS AND DEPLOYMENT TRADE-OFFS VISIBLE.

Design choice	A useful question
Data and splits	Are test items leaked, duplicated, or too similar to training data?
Metrics and judges	Does the score track the intended capability? Do judges agree with humans?
Compute and tools	Were resolution, tokens, samples, tools, and latency controlled or reported?
Robustness and change	Do conclusions hold across subsets, shifts, and benchmark versions?

Video generation: test appearance, instruction following, and physics.

Four questions to bring to the reading

FOR EVERY PAPER THIS SEMESTER

- 01 What exactly is the claim?
- 02 What evidence supports it, and what would refute it?
- 03 Are the baselines and the compute budget fair?
- 04 What does it not show? Where would it break?

Next week:
Post-training and RL